LED Control System

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Introduction

- This project aims to build an efficient LED control system using an Arduino microcontroller. It helps in understanding basic circuit building and embedded programming.

Objective

- To develop a simple, low-cost system that allows LEDs to be controlled using code and demonstrates basic logic flow in embedded systems.

Components Used

- Arduino Uno
- LEDs (Red, Green, Blue)
- Resistors (220 ohms)
- Jumper Wires
- Breadboard
- USB Cable for Power

Working Principle

- The Arduino sends output signals to the LEDs through digital pins. Each LED lights up one after another with a time delay, creating a sequence pattern.

Circuit Description

- Three LEDs are connected to digital pins 9, 10, and 11 respectively. Each pin is programmed to turn the LEDs on and off in a cycle with a 1-second delay.

Microcontroller Used

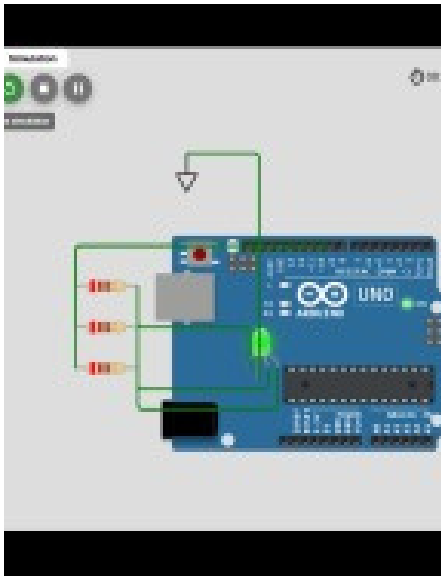
- We use the Arduino Uno microcontroller, which is an open-source electronics platform based on easy-to-use hardware and software. It is ideal for prototyping.

Flow Chart

- Start -> Initialize pins -> Turn ON Red -> Delay -> Turn OFF Red, Turn ON Green -> Delay -> Turn OFF Green, Turn ON Blue -> Delay -> Repeat

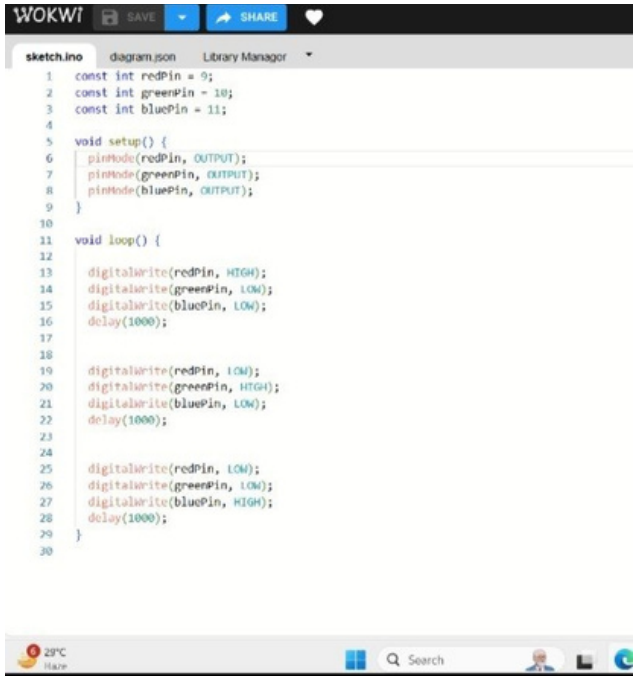
Block Diagram

- Refer to the block diagram where the Arduino is connected to 3 LEDs with appropriate resistors. It shows the flow from code to hardware.



Program Code

- See the embedded code. It cycles through the LEDs one by one using `digitalWrite()` and `delay()` functions in the Arduino IDE.



Implementation

1. Connect the LEDs to the Arduino board. 2. Open Arduino IDE and upload the program.
3. Power the board and observe the LED sequence.
4. Debug if needed.

Advantages

- Low Power Consumption
- Easy to Program and Use Inexpensive
- Components Scalable for Larger Projects

Applications

- Smart Home Lighting
- Indicator Systems
- School and College Projects
- Industrial Light Controls

Challenges Faced

- Initially faced wiring issues and incorrect pin mappings. Solved by reviewing pin definitions and ensuring correct connections.

Future Scope

- Can be expanded with IoT integration using Wi-Fi modules like ESP8266, or controlled using mobile apps and voice assistants like Alexa.

Conclusion

This project enhanced our understanding of embedded systems. We thank our mentor Mr. Santosh and JJ College for the guidance and support.